Time as a Sequence:

From Gluon Plasma to the Non-Singular Star

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ABSTRACT

This monograph collects three papers on one hypothesis: time is not a static background coordinate, it is the sequence itself – an ordered chain of phase transitions and structural transformations of matter, from gluon–quark plasma to chemistry and back. Each stage of the sequence has its own tempo. In this hypothesis, gravity can reverse the sequence, which means it can slow time down, stop it, and even reverse it. The proposed mechanism is the accumulation of gluons. The end of the sequence is not a singularity and not an event horizon, but a stable Non-Singular Star (a star without infinite collapse) formed from one ideal package of mass (approx. 2.8 $M_{\odot}$).

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PART I

Time is the Sequence Itself – Slowik Hypothesis, Formula 5+1+5

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1.1 Core Definition

Time is the sequence itself – from gluon to chemistry and back. Each stage has its own tempo: free neutron 15 minutes (pure magnetism); in a neutron star gravity reverses beta decay ($p + e^- \rightarrow n + \nu$) [10], so the neutron there is stable, proton eternal (electro/magnetism). In this hypothesis gravity can reverse this sequence, which would mean that it can slow time down, stop it, and even reverse it. The zero point is the gluon–quark plasma [11, 12] – where the sequence begins and ends. In this picture time has a frequency, gravity controls it, and the proposed mechanism is the accumulation of gluons.

1.2 Full Chain – Frequency of Time

Stage	Element of the sequence	Remark
1	Higgs field	beginning and end
2	Gluon–quark plasma	zero point, deconfinement, zero time
3	Neutron	magnetism, down quarks dominate, 15 min lifetime (free neutron), world without light
4	Beta decay	down → up, W/Z, positron/electron
5	Proton	electro/magnetism, eternal, up quarks
6	Electromagnetism	
7	Photon, light	
8	Electron orbit	needs a magnetic field to orbit
9	Chemistry	
10	Biology	
11	Life	
12	Star	gravity compresses matter; synthesis (fusion) builds an increasingly compact structure [9]
13	White dwarf	end state of a star such as the Sun after it burns out; electron-degenerate matter, atoms crushed [6]
14	Neutron star	stop time, creation, magnetism protects; gravity reverses beta decay, neutron stable
15	Black hole	Package Formula
16	Accumulation of gluons	gravity reverses the sequence
17	Back to gluon plasma	

PACKAGE FORMULA

> 1 Package = 2.8 $M_{\odot}$
>
> $R_s = N \times 8.27$ km
>
> $R_{core} = 15.8$ m
>
> $B_n = 0$ – firmware holds all layers
>
> $S_n = 2F_n$
>
> $S_n = E_{core} + E_{magnetic}$
>
> **5+1+5 – NOT STEP BY STEP, ONE PACKAGE**

Note on the formula. $R_s = N \times 8.27$ km is not an independent formula. It is a simplified form of the Schwarzschild radius $R_s = 2GM/c^2$ [4, 5], with $M = N \times 2.8\ M_{\odot}$ (N = number of packages): $2GM_{\odot}/c^2 \approx 2.95$ km, and 2.95 km × 2.8 ≈ 8.27 km. Only the choice of 2.8 $M_{\odot}$ as one "package" is the author's hypothesis; in the model tests [18] the mass is varied over 2.8–3.2 $M_{\odot}$. $R_{core} = 15.8$ m is a heuristic scale GMm_0/X ($m_0 \approx 0.94$ GeV/c², X = Higgs vacuum expectation value), not the radius of the object: in the tests the physical shell radius is a ≈ 9.8–10.5 km for 2.8 $M_{\odot}$, with exterior compactness $2GM/(c^2a) \approx 0.785$–0.844 [18].

1.3 Gravity and Time

Suppose, as a hypothesis, that time has an intrinsic frequency. In a gravitational field spacetime curvature changes the path and rate associated with a clock. If this temporal behavior could be represented by a frequency, controlling that frequency allows to control the rate at which time is experienced – slowing it progressively and, in the limiting case, approaching a state in which temporal evolution effectively stops. Gravity reveals a deeper relationship between spacetime geometry and temporal frequency. Manipulating that relationship provides a mechanism for controlling the rate of time.

Proposed mechanism = accumulation of gluons. Gravity feels the gluon is active – which is why it gravitates.

1.4 Biology – Separate Rhythm

Biology cannot be reversed or accelerated – because it was built in biological reactions. It has its own rules, separate from gravitational time. But at large mass, for an external observer, it can slow down. That's the limit: biology is not physics – it's built on physics, but it has its own rhythm. Chemistry → biology → life is a natural emergent mechanism, universal across the universe.

1.5 Magnetism First

Magnetism was there first – both proton and neutron have it. It contributes to the tensor just like the electric part does, because it emerges when quarks are captured by the gluon. The "electro" part only appears after beta decay. The gluon begins its action when mass is given by the Higgs boson [13, 14]. Without magnetism, there would be no proton + electron – because there would be no magnetic field for the electron to orbit around.

Magnetism → beta decay → electro → electro/magnetism. Two worlds, one interaction – from the gluon. Holds quarks together.

PART II

Time as a Sequence of Transitions and Mass Regulation Mechanism: From Hydrogen Cloud to Non-Singular Star

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2.1 Abstract

This part presents a novel interpretation of the nature of time and the evolution of gravitational objects, shifting from the concept of a static background coordinate to a sequential framework. Time is defined here as an ordered sequence of phase transitions and structural transformations of matter. I introduce the concept of an optimal (ideal) primary hydrogen cloud mass whose gravitational collapse leads to precise transitions down to the core level. Similar to the helium fusion process [9], where a mass defect results in the ejection of excess energy, I hypothesise that the gravitational system strives to achieve an ideal, balanced "packet" (approx. 2.8 $M_{\odot}$; the tests in [18] cover 2.8–3.2 $M_{\odot}$), discarding excess matter during a supernova event. This mechanism prevents the formation of unphysical singularities, instead forming a stable Non-Singular Star structure.

2.2 Introduction: Time as a Sequence of Transitions

In traditional physical models, time is treated as an independent, continuous coordinate. I propose an alternative perspective in which time is simply a sequence – a measure of occurring phase transitions and structural transformations of matter. Without an active sequence of interactions and energy exchange, the concept of the passage of time loses its physical meaning.

This sequence is not the flow of an abstract stream, but an order of events driven by matter's drive toward equilibrium.

2.3 The Role of the Hydrogen Cloud and the Optimal Evolutionary Path

The gravitational evolution process depends on the initial amount of matter in the hydrogen cloud:

- **Insufficient mass.** If the initial amount of hydrogen is too small, gravity cannot force deeper phase transitions. The process halts at earlier stages of degeneracy, leading to the formation of a white dwarf, pulsar, or magnetar [6, 7, 8].
- **Excess mass.** When the mass of a collapsing star is very large, gravity and synthesis processes accelerate drastically. However, the system does not strive for infinite concentration at a point (singularity), but triggers a regulatory mechanism.

2.4 Mass Excess Rejection and the Analogy with Mass Defect

Similarly to nuclear physics, where hydrogen fusion into helium (or particle production) involves a mass defect – with the excess being emitted because the final structure must maintain precise proportions – the same occurs on a macroscopic scale:

- **The necessity of packet closure.** In this hypothesis, for a stable, primordial object (Non-Singular Star) to form, the system possesses a well-defined base mass (the standard "packet", approx. 2.8 $M_{\odot}$ as the reference value; the tests in [18] vary the mass over 2.8–3.2 $M_{\odot}$).
- **The role of supernovae.** In the case of excessive stellar mass, a supernova explosion acts as a natural mechanism rejecting excess matter. The star sheds what disturbs the balance until the remainder achieves an ideal, closed measure capable of stabilization.

2.5 Conclusions

Defining time as a sequence of transitions and introducing the mechanism of mass regulation through excess rejection is proposed here as a way to address dilemmas in modern astrophysics. In this picture, instead of leading to infinite singularities, gravity and subatomic interactions cooperate as a packaging mechanism that forms stable objects without event horizons; static configurations of this kind are tested in [18].

PART III

On the Dynamic Nature of Time and the Elimination of Event Horizons: The "Non-Singular Star" Hypothesis and Motion-Exchange Framework

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3.1 Abstract

This part presents a theoretical framework addressing fundamental anomalies in modern gravitational physics, specifically the paradoxes of singularities and event horizons in black hole theory. I introduce the concept of time not as an immutable, abstract background coordinate, but as a direct emergent property governed by physical movement, energy exchange, and multi-scale interactions. Building upon this, I formalize the "Non-Singular Star" model – an alternative macroscopic object configuration stabilized by anisotropic stresses in a thin shell [18], entirely circumventing gravitational collapse to a mathematical singularity. Furthermore, I explore the integration of subatomic interactions (such as gluon dynamics) with macroscopic gravitational fields through a unified exchange mechanism.

3.2 Introduction and Motivation

Modern theoretical physics faces severe conceptual roadblocks at the intersection of general relativity and quantum mechanics, most notably the emergence of infinite densities (singularities) and event horizons within gravitational collapse models. Traditional paradigms treat space and time as a passive stage upon which physical events unfold.

However, a rigorous re-evaluation suggests that time is intrinsically tied to the active processes of the universe. To resolve these contradictions, I propose a framework where the absence of singularities is guaranteed by active energy-exchange mechanisms, scaling

coherently from quantum structures to cosmological entities.

3.3 Time as an Emergent Property of Motion and Exchange (Motion-Breath-Exchange Framework)

In standard formulations, time (t) is treated as an independent scalar parameter. In the proposed hypothesis:

- **Time is a derivative of dynamics.** Time does not exist independently of physical processes; rather, it is the quantitative measure of motion, transformation, and energy exchange between systems.
- **Scale bridging.** The micro-scale dynamics (quantum fluctuations, particle-level exchanges) dictate the macro-scale progression of time. The local “tempo” of time scales proportionally with the intensity of energy transfer within a given bound system.
- **Elimination of static parameters.** By redefining time as an active exchange, we remove the mathematical artifacts that lead to frozen coordinates or infinite dilations at hypothetical horizons.

3.4 The “Non-Singular Star” Model: An Alternative to Black Holes

To resolve the physical impossibility of infinite density singularities, I introduce the Non-Singular Star architecture:

- **Absence of event horizons.** Unlike classical black holes bounded by an event horizon from which no information escapes, the Non-Singular Star has a Schwarzschild vacuum exterior and a physical surface at a radius larger than its Schwarzschild radius, so signals from the surface can reach the exterior; a possible observational signature is gravitational-wave echoes [18].
- **Pressure equilibrium.** Gravitational collapse is halted prior to singularity formation by anisotropic stresses: a thin shell with tangential pressure carries the weight of the object [15, 16, 17], and static, radially stable configurations exist for exterior compactness $2GM/(c^2a)$ between 0.785 and 0.844 [18].
- **Subatomic foundations.** The structural integrity of a macro-object in this phase is maintained by scaling up fundamental interactions, effectively coupling strong-force characteristics and gluon-mediated dynamics with gravitational bounding. The microphysical origin of the shell matter remains an open question [18].

3.5 Mathematical Notations and Packet-Based Interactions

To describe the transition between discrete quantum states and continuous field descriptions, I incorporate a packet-based mathematical formalism:

- **Interaction packets.** Energy and momentum transfers are treated as discrete packets rather than continuous infinite streams, mitigating self-energy divergences.
- **Field coherence.** The mathematical treatment avoids division-by-zero anomalies characteristic of standard Schwarzschild metrics [4] by introducing lower-bound length and density limits enforced by subatomic pressure thresholds.

3.6 Conclusions

The hypothesis presented here offers a singularity-free pathway for understanding extreme gravitational phenomena. By redefining time as an emergent property of motion-exchange and introducing the Non-Singular Star model, we aim to bridge the gap between quantum realities and astrophysical structures without resorting to unphysical infinities. No compact object of this kind has been identified observationally; the tests in [18] give no observable predictions beyond possible gravitational-wave echoes.

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